

Device Fabrication/ Fabrication Of Chips And Microprocessors:

The fabrication of a device involves a multi-step procedure for manufacturing and assembling numerous technical components. In semiconductor engineering, this procedure is applied to the production of chips, integrated circuits (ICs), and microprocessors, where carefully controlled physical and chemical processes are used to build highly compact and efficient electronic circuits on silicon wafers.

Basic Steps in Chip Fabrication

Fabrication of chips is an incredibly precise process that turns raw silicon into a functional brain for an electronic device Steps for Fabrication of chips:

1. Silicon Wafer Preparation
2. Deposition & Coating
3. Photolithography
4. Etching & Ion Implantation

Q. Comment on deposition techniques -

1. PVD
2. CVD
3. AVD

1.Physical Vapor Deposition (PVD)

Physical Vapor Deposition (PVD) is a widely used thin film deposition technique in the manufacturing of devices. PVD involves the *physical ejection of material from a source (target) and its subsequent deposition onto a substrate*, typically under vacuum conditions.

A. Sputtering Deposition

Sputtering deposition is a PVD method commonly employed in the optoelectronics industry. In this process, a target material is bombarded with *high-energy particles, typically ions, causing the target atoms to be ejected and deposited onto the substrate*. Sputtering deposition allows for the deposition of a wide range of materials, including metals, alloys, and ceramics, making it a versatile technique for the fabrication of optoelectronic devices.

B. Evaporation Deposition

Evaporation deposition is another PVD technique used in the manufacturing of electronic devices. In this process, the *source material is heated, typically using an electron beam or resistance heating, causing it to evaporate and condense onto the substrate*. Evaporation deposition is particularly well-suited for the deposition of high-purity thin films, as it allows for excellent control over the deposition rate and film thickness.

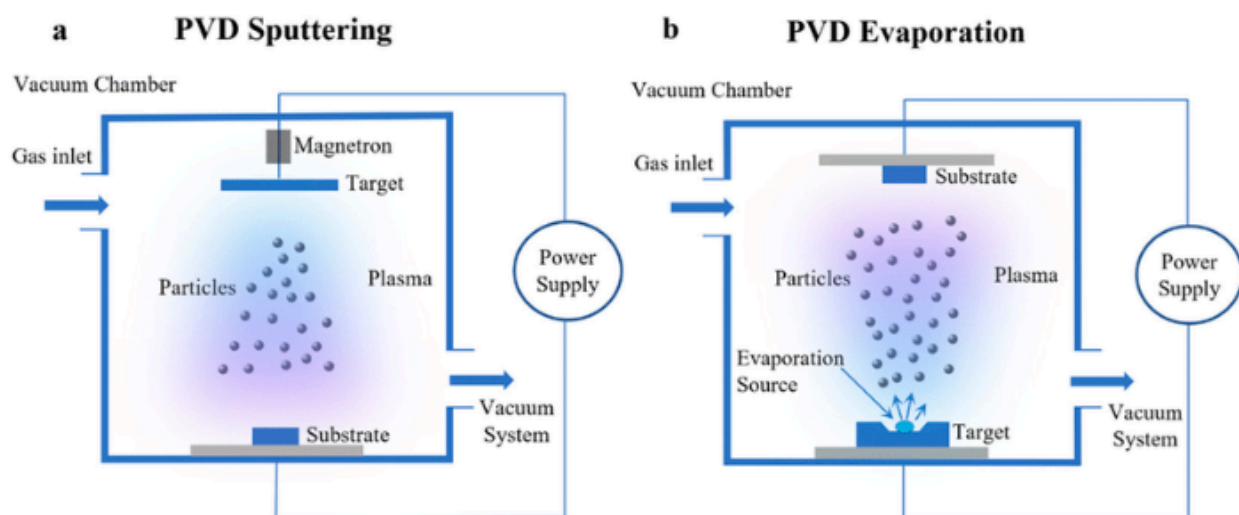


FIGURE: PVD deposition coating methods: (a) sputtering, (b) evaporation.

2. Chemical Vapor Deposition (CVD)

Chemical Vapor Deposition (CVD) is another thin film deposition technique that is widely used in the manufacturing of electronic devices. In **CVD**, a substrate is exposed

to one or more volatile precursor chemicals, which react and/or decompose on the substrate surface to produce the desired thin film material. This process allows for the precise control of film composition, thickness, and uniformity, making it a crucial tool in the fabrication of high-performance optoelectronic components.

CVD working principle:

Chemical vapor deposition processes can be defined as any process in which a thin solid film is formed on a substrate by the ***surface-mediated reaction of adsorbed precursors from the gas phase***. The reactive nature of CVD processes distinguishes them from physical processes such as evaporation and sputtering that are employed in PVD. The term "surface-mediated" refers to the fact that the solid film is formed by a heterogeneous reaction occurring at the substrate surface.

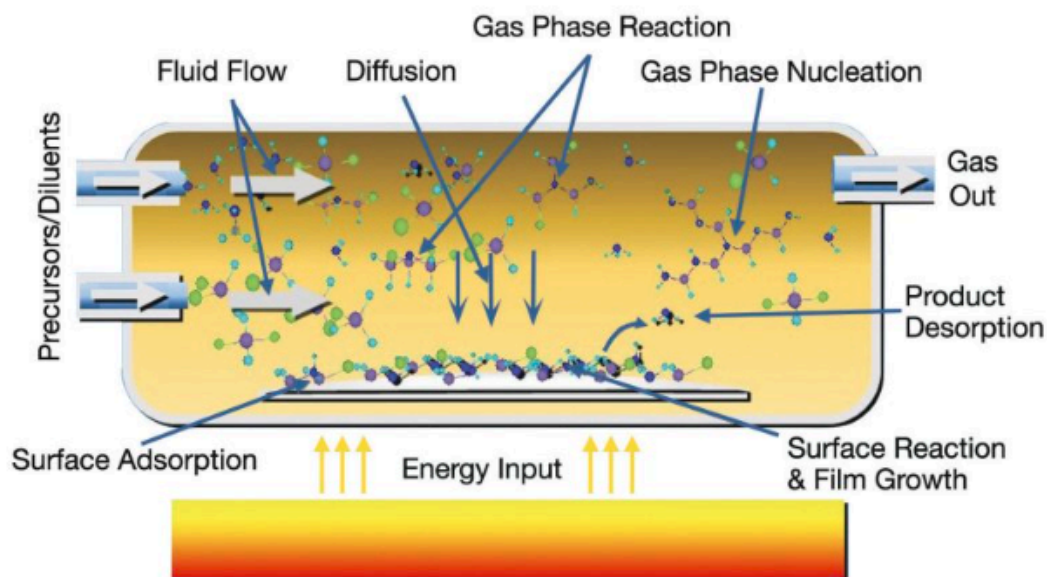


Figure: Aspects of CVD

Above Figure shows a schematic that can help in understanding the different processes in a CVD reactor. Chemical vapor deposition processes can be broken down into a number of discrete steps:

First the precursor chemicals must be fed into the CVD reactor. Once in the reactor, precursor molecules must be transported to the substrate surface, usually through a combination of fluid transport and diffusion. Once on the surface, the precursor molecule must remain there long enough to react. After reaction occurs, the product thin film atom must remain on the surface while the by-product molecules must desorb from the substrate surface to make room for more incoming precursor molecules.

Low-Pressure CVD

Low-Pressure CVD (LPCVD) is a variant of the CVD process that operates under reduced pressure conditions. By lowering the pressure, **LPCVD can improve the quality and uniformity of the deposited thin films, as well as increase the deposition rate.** This technique is commonly used for the deposition of polycrystalline silicon, silicon nitride, and other materials critical to the production of optoelectronic devices.

Plasma-Enhanced CVD

(Note: Plasma is the fourth state of matter—distinct from solid, liquid, or gas—consisting of highly ionized gas with free-moving electrons and ions.)

Plasma-Enhanced CVD (PECVD) is another important CVD technique that utilizes a plasma to activate and decompose the precursor gases. The plasma provides the necessary energy to *facilitate the chemical reactions at lower substrate*

temperatures, making it a versatile option for depositing thin films on temperature-sensitive materials. PECVD is widely used in the production of amorphous silicon, silicon nitride, and other thin film coatings for optoelectronic applications.

Importance

A) Chemical Vapor Deposition (CVD) for devices is a manufacturing process that deposits high-purity, high-performance solid materials onto a substrate (wafer) by triggering chemical reactions in a vapor phase. It is essential for creating thin films in semiconductors, MEMS, and electronics, specifically depositing dielectric layers, metals, and carbon nanostructures

B) Typical films produced by CVD processes include:

(Note: Epitaxy is a fabrication process used in semiconductor manufacturing to grow a thin, single-crystal film on a crystalline substrate, ensuring the new layer matches the crystal orientation of the underlying material. It enables precise, high-purity, and defect-free material, crucial for advanced microchips, solar cells, and LED devices.)

- Epitaxial Silicon (Epitaxial silicon (EPI) is a high-purity, monocrystalline silicon layer grown on a polished silicon substrate, typically via chemical vapor deposition (CVD) at temperatures around 1100°C)
- Epitaxial Compound Semiconductors
- Polycrystalline Silicon
- Dielectric Thin Films

- Silicon Dioxide (including P- and B-doped Oxides)
- Aluminum Oxide
- Low-k Dielectrics (SiO-F, DLC, amorphous C-F) (Low-k dielectrics are insulating materials with a low dielectric constant, $k < 3.9$ the value for silicon dioxide) used in semiconductor manufacturing to separate electrical interconnects, reducing signal crosstalk and latency while lowering heat dissipation)
- Silicon Nitride
- Titanium Nitride
- Tantalum Nitride
- Transition Metal Oxides (i.e. TiO_2 , ZrO_2 , Hf_2O_5 , Ta_2O_5)
- Metals (Al, W, Cu)
- Silicides (WSi_x , CoSi_x)

3. Atomic Layer Deposition (ALD)

In the world of thin film deposition, Atomic Layer Deposition (ALD) has emerged as a groundbreaking technique that is revolutionizing the manufacturing of optoelectronic devices. Unlike traditional thin film deposition methods, ALD is a self-limiting, sequential process that involves the exposure of a substrate to **alternating precursor gases, which react to form a single atomic layer of the desired material.**

The precision and control offered by ALD make it an invaluable tool for nanoscale engineering, where the ability to precisely control the thickness and composition of thin films is crucial. This precise control, combined with the ability to deposit high-quality,

conformal films on complex surfaces, has made ALD a go-to choice for the fabrication of cutting-edge optoelectronic devices, ranging from solar cells to semiconductor lasers.

As the demand for ever-smaller and more efficient optoelectronic devices continues to grow, the role of ALD in the manufacturing process has become increasingly important. By enabling the deposition of ultra-thin, high-quality films with exceptional uniformity and conformality, ALD has become a key enabler of the ongoing advancements in the optoelectronics industry, paving the way for the development of innovative new technologies that push the boundaries of what is possible.

Working Principles:

Atomic layer deposition is similar to LPCVD except that the chemical process is broken down into steps that isolate different adsorption and reaction steps to have self-limiting reactions. The process employs separate pulses of precursors and reactants that pass sequentially through the process chamber. Figure 1 illustrates the ALD process. With *the substrate in the process chamber and under high vacuum, an initial precursor is introduced into the chamber. The molecular character of the precursor is such that it will form a chemically-bound monolayer on the substrate surface (Step 1). Any layers beyond the monolayer are only bound by physisorption forces which are weak enough to allow any precursor other than that in the monolayer to be pumped away under high vacuum. Once the monolayer is present on the substrate, the chamber is re-evacuated and purged to remove any excess precursor* (Step 2). Next, a reactant is introduced to the process chamber (Step 3). It reacts with the monolayer material to form the desired

compound on the substrate surface (Step 4). By-products of this reaction are pumped away.

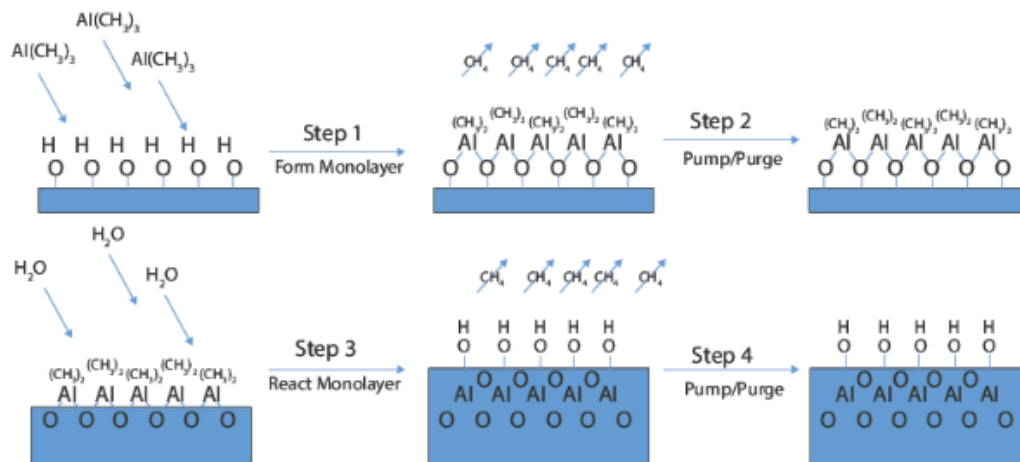


Figure 1. Steps in an ALD process.